Data sheet



HPE Networking SRX1600 Firewall

Product Description

The HPE Networking SRX1600 Firewall is a high-performance, next-generation firewall (NGFW) designed to safeguard your enterprise campus edge, data center edge, and branch offices. It also supports roaming, SD-WAN large branch, and SD-WAN secure hub use cases. Combining industry-leading security efficacy and carrier-grade routing with state-of-the-art switching, this platform delivers robust network security, effective threat protection, and comprehensive automation and mitigation capabilities.



Figure 1. HPE Networking SRX Firewalls have achieved the highest scores in security effectiveness by CyberRatings and NetSecOpen

As network architectures become more distributed and decentralized, HPE Networking SRX Firewalls ensure seamless integration with other HPE and third-party networking platforms. At the same time, the NGFWs facilitate architectural transformation, taking organizations from on-premises to hybrid cloud environments seamlessly and cost-effectively. SRX series are the first to implement industry-standard Ethernet VPN (EVPN) Type 5 and Virtual Extensible LAN (VXLAN) protocols within data center environments, enabling the SRX1600 to act as a secure, fabric-aware leaf in the data center spine-leaf architecture.

The SRX1600 participates in HPE Networking connected security distributed services architecture, enabling organizations to scale horizontally and elastically, simplifying the operational management of large-scale firewall networks. With this architecture, several SRX1600 platforms can work together as a single large logical firewall to provide higher performance and scale security.

The SRX1600 is powered by the industry-leading HPE Networking Junos Operating System that underpins and helps secure the world's largest mission-critical enterprise and service provider networks. It is managed by HPE Networking Security Director, the unified management experience of HPE that connects the organization's current deployments with future architectural rollouts. HPE Networking Security Director uses a single-policy framework, enabling consistent security policies across any environment and expanding zero trust to all parts of the network, from the edge into the data center. This provides unbroken visibility, policy configuration, administration, and collective threat intelligence all in one place.

Product Overview

The firewall's role must expand as data centers evolve from traditional architectures to distributed ones. Rather than being a perimeter technology, firewalls need to be part of a security fabric woven throughout the network. A security fabric ensures that security is maintained at every point of connection. HPE Networking connected security distributed services architecture, managed by HPE Networking Security Director, offers a high-performance, scalable, and easy-to-manage firewall solution to secure today's distributed data centers. HPE Networking SRX1600 Firewall is integral to this new architecture and empowers organizations to operationalize security across their networks. This 1U, power-efficient firewall features built-in zero trust, Ethernet VPN-Virtual Extensible LAN (EVPN-VXLAN) fabric integration, and AI-predictive threat prevention to secure your network.

Architecture and key components

HPE Networking SRX1600 Firewall hardware and software architecture provides cost-effective security in a compact, scalable 1U form factor. Purpose built to protect network environments, the SRX1600 incorporates multiple security services and networking functions on top of HPE Networking Junos Operating System, providing highly customizable threat protection, automation, and integration capabilities. Best-in-class advanced security capabilities on the SRX1600 are offered in the data center, enterprise campus, and regional headquarters deployments with IMIX traffic patterns.

Built-In zero trust

The SRX1600 features built-in zero trust device capabilities to increase trust and streamline operations, including an embedded Trusted Platform Module (TPM) 2.0 and cryptographically signed device ID. The SRX1600 supports RFC-compliant secure zero touch provisioning (sZTP) to efficiently, expediently, and remotely deploy products in your network. Additionally, the SRX1600 supports MACsec at wire speed, ensuring data integrity and confidentiality.

HPE Networking connected security distributed services architecture

The SRX1600 is part of HPE Networking connected security distributed services fabric, revolutionizing data center security. With HPE Networking connected security distributed architecture, firewall performance can scale horizontally by interconnecting traffic forwarding and security services across multiple locations. The HPE solution also provides automated failover and backup nodes for forwarding and inspection components. In addition to redundancy and load balancing, the HPE Networking connected security distributed architecture simplifies how large-scale data center firewall networks are managed and operated.

Regardless of how many firewall engines are added across the various form factors, they can all be managed as one logical unit. The centralized management eliminates the complexity that has been an unintended consequence of a traditional scale-out approach.

Features and benefits

Business requirement	Feature/solution	SRX1600 advantages
High performance	Hardware-accelerated encryption/decryption	— Offloads CPU-intensive encryption/decryption tasks — Improves performance for SSL and IPSec
High-quality end user experience	Application visibility and control	— Updates application continuously and decodes custom applications — Controls and prioritizes traffic based on application and user role — Inspects and detects applications inside SSL-encrypted traffic, including Web and SaaS
Advanced threat protection	NGFW Services: IPS, antivirus, antispam, web filtering, HPE Networking Security Director cloud: sandboxing, encrypted traffic insights, security intelligence	— Prevents exploits with 99.9% effectiveness; signatures update in real time — Protects against known malware and malicious Web and DNS traffic — Sandboxing for unknown malware across multiple OS types, including iOS, Windows, Android™, and CentOS — Delivers threat intelligence in an open platform to accommodate third-party and custom threat feeds — Detects threats hidden inside encrypted traffic without decrypting
Zero-day protection	AI-predictive threat prevention	— Predicts and prevents malware at line rate by using AI to identify threats from packet snippets effectively — Eliminates patient-zero infections — Auto-generates protective signatures that remain active for the full attack lifecycle, keeping the network safe from subsequent attacks

Business requirement	Feature/solution	SRX1600 advantages
Secure data transactions	HPE Networking SRX Secure Connect: IPSec VPN, remote access/ SSL VPN	— Provides high-performance IPSec VPN with a dedicated crypto engine — Offers diverse VPN options for various network designs, including remote access and dynamic site-to-site communications — Simplifies large VPN deployments with auto-VPN — Includes hardware-based crypto acceleration — Ensures secure and flexible remote access SSL VPN
Advanced networking services	Routing, secure wire	— Supports carrier-class advanced routing and quality of service (QoS)
Security embedded into the data center fabric	EVPN-VXLAN (EVPN Type 5 routes)	— Enhances tunnel inspection for VXLAN encapsulated traffic with Layers 4-7 security services — Eases operations with Type 5 support through BGP — Does not require decapsulation for EVPN-VXLAN traffic
Reliability	Chassis cluster, MNHA redundant power supplies	— Provides stateful configuration and session state synchronization — Supports active/active and active/backup deployment scenarios — Offers highly available hardware with redundant power supply unit (PSU) and fans
Easy to manage and scale	HPE Networking Security Director cloud, on-box GUI	— Provides centralized management via the unified management experience of HPE, including zero-touch provisioning (ZTP), unbroken visibility, intelligent rule placement, and simplified policy configuration and automation — Supports Network Address Translation (NAT) and automated IPSec VPN deployments via wizards — Supports on-box GUI
Built-in zero trust capabilities	DevID with TPM 2.0 Module	— Verifies the devices' trust posture easily — Provides cryptographically signed device ID that supports RFC8572-compliant sZTP for hardware and software attestation — Mitigates the risks of supply chain attacks
Low TCO	HPE Networking Junos Operating System	— Integrates routing and security capabilities into a single device — Reduces OpEx with Junos OS automation capabilities — Automates integration with other devices running Junos OS, such as HPE Networking routers, HPE Networking EX Switches, and HPE Networking QFX Switches

Software specifications

Firewall services

— Stateful firewall services
— Zone-based firewall
— Screens and distributed denial of service (DDoS) protection
— Protection from protocol and traffic anomalies
— Unified Access Control (UAC)
— User role-based firewall
— SSL inspection
— Integration with HPE Mist Networking Access Assurance

Carrier-grade network address translation (CGNAT)

— Carrier-grade Network Address Translation (large-scale NAT)
— IPv4 and IPv6 address translation NAT44, NAPT44, NAT66, NAPT66, NAT64, NAT46
— Static and dynamic 1-1 translation
— Source NAT with Port Address Translation (PAT)
— Destination NAT with Port Address Translation (PAT)
— Persistent NAT (EIM/EIF)
— Port Block Allocation (PBA)
— Deterministic NAT (DetNAT)
— Port overload
— Twice-NAT44
— DS-lite and Port Control Protocol (PCP)

VPN features

— Tunnels: Site-to-site, hub and spoke, dynamic endpoint, AutoVPN, ADVPN, Group VPN (IPv4/ IPv6/ dual stack)
— HPE Networking SRX Secure Connect: Remote access/SSL VPN
— Configuration payload: Yes
— IKE encryption algorithms: Prime, 3DES-CBC, AEC-CBC, AES-GCM, Suite B
— Authentication: Pre-shared key and public key infrastructure (PKI) (X.509)
— IPSec: Authentication Header (AH)/Encapsulating Security
— Payload (ESP) protocol
— IPSec authentication algorithms: hmac-md5, hmac-sha-196, hmac-sha-256
— IPSec encryption algorithms: Prime, DES-CBC, 3DES-CBC, AEC-CBC, AES-GCM, Suite B
— Perfect forward secrecy, anti-reply
— Internet Key Exchange: IKEv1, IKEv2
— Monitoring: Standard-based dead peer detection (DPD) support, VPN monitoring
— VPNs GRE, IP-in-IP, and MPLS

High Availability Features

— Virtual Router Redundancy Protocol (VRRP)–IPv4 and IPv6
— Stateful high availability: Dual box clustering
  • Active/passive
  • Active/active
  • Configuration synchronization
  • Firewall session synchronization
  • Device/link detection
  • In-Service Software Upgrade (ISSU)
  • IP monitoring with route and interface failover
  • BFD monitoring
— Chassis cluster HA and Multinode HA (MNHA)

Application security services[1]

— Application visibility and control
— Application quality of service
— Advanced/application policy-based routing (APBR)
— Application quality of experience (AppQoE)
— Application-based multipath routing
— User-based firewall

Threat Defense and Intelligence Services[1]

— Intrusion prevention system
— AI-predictive threat prevention
— Antivirus
— Antispam

[1] Offered as an advanced security subscription license

— Category/reputation-based URL filtering
— SSL proxy/inspection
— Protection from botnets (command and control)
— Adaptive enforcement based on GeoIP
— HPE Networking Advanced Threat Prevention, a cloud-based SaaS offering to detect and block zero-day attacks
— Adaptive threat profiling
— Encrypted traffic insights
— Security intelligence
— HPE Networking Advanced Threat Prevention virtual appliance, a distributed, on-premises advanced threat prevention solution to detect and block zero-day attacks

Routing protocols

— IPv4, IPv6, static routes, RIP v1/v2
— OSPF/OSPF v3
— BGP with route reflector
— IS-IS
— Multicast: Internet Group Management Protocol (IGMP) v1/v2, Protocol Independent Multicast (PIM) sparse mode (SM)/source-specific multicast (SSM); Session Description Protocol (SDP), Distance Vector Multicast Routing Protocol (DVMRP); Multicast Source Discovery Protocol (MSDP), reverse path forwarding (RPF)
— Encapsulation: VLAN, Point-to-Point Protocol over Ethernet (PPPoE)
— Virtual routers
— Policy-based routing, source-based routing
— EVPN-VXLAN (EVPN Type 5 route)
— Equal-cost multipath (ECMP)

QoS features

— Support for 802.1p, DiffServ code point (DSCP), EXP
— Classification based on VLAN, data-link connection identifier (DLCI), interface, bundles, or multifield filters
— Marking, policing, and shaping
— Classification and scheduling
— Weighted random early detection (WRED)
— Guaranteed and maximum bandwidth
— Ingress traffic policing
— Hierarchical shaping and policing
— Virtual channels

Switching Features

— ASIC-based Layer 2 forwarding
— MAC address learning
— VLAN addressing and integrated routing and bridging (IRB) support
— Link aggregation and LACP
— Link Layer Discovery Protocol (LLDP) and Link Layer Discovery Protocol-Media Endpoint Discovery (LLDP-MED)
— STP, RSTP, MSTP
— Multiple VLAN Registration Protocol (MVRP)
— 802.1x authentication
— MACsec

Network services

— Dynamic Host Configuration Protocol (DHCP) client/server/relay
— Domain Name System (DNS) proxy, dynamic DNS (DDNS)
— HPE real-time performance monitoring (RPM) and IP monitoring
— Flow-monitoring

Advanced routing services

— MPLS (RSVP, LDP)
— Circuit cross-connect (CCC), translational cross-connect (TCC)
— L2/L2 MPLS VPN, pseudo-wires
— Virtual private LAN service (VPLS), next-generation multicast VPN (NG-MVPN)
— MPLS traffic engineering and MPLS fast re-route

Management, automation, logging, and reporting

— SSH, Telnet, SNMP-MIBS, Traps
— Smart image download
— HPE CLI and Web UI, NetCONF, XML APIs, RMON
— HPE Networking Security Director cloud
— Python
— Junos events, commit and OP scripts
— Application and bandwidth usage reporting
— Debug and troubleshooting tools

Hardware specifications

Table 3. HPE Networking SRX1600 Firewall hardware specifications

Specifications	SRX1600
Connectivity	
Onboard ports	16 x 1GbE 10/100/1000 BASE-T
Onboard small form-factor pluggable plus (SFP+) transceiver ports	4 x 1GbE/10 GbE SFP+ 2 x 1GbE/10 GbE/25 GbE SFP28
Out-of-band (OOB) management ports	1 x 1GbE G (RJ-45)
Dedicated high availability (HA) ports	2 x 1GbE SFP
Console	1 (RJ-45)
USB 3.0 ports (Type A)	1
Storage	
Storage (SSD)	1 x 120 GB
Dimensions and power	
Form factor	1U
Size (W x H x D)	17.28 x 1.74 x 18.20 in (43.89 x 4.42 x 46.23 cm)
Weight (device and PSU)	Chassis with two AC power supplies: 15.7 lb (7.1 kg) Chassis with two DC power supplies: 15.9 lb (7.2 kg) Chassis with package for shipping: 32.8 lb (14.9 kg)
Redundant PSU	1+1
Power supply	2 x 450 W AC PSU redundant 2 x 650 W DC PSU redundant
Average heat dissipation	1 x DC PSU (40V): 487.9 BTU/h 2 x DC PSU (40V): 498 BTU/h 1 x AC PSU (110V): 467.5 BTU/h 1 x AC PSU (230V): 445.3 BTU/h 2 x AC PSU (110V): 510 BTU/h 2 x AC PSU (230V): 501.6 BTU/h
Maximum current consumption	2A (for 110 V AC PSM) 1A (for 230 V AC PSM) 4.7A (for -40 V DC PSM)
Maximum inrush current	50A for 1 cycle of AC (AC PSM) 40A-pk (DC PSM)
Environment and regulatory compliance	
Acoustic noise level	58 dB
Airflow/cooling	Front to back
Operating temperature	32°F to 104°F (0°C to 40°C at 6000 ft altitude)
Operating humidity	5% to 90% non-condensing
Meantime between failures (MTBF)	Over 100,000 hours (12 years)
FCC classification	Class A
RoHS compliance	RoHS 6
Performance and scale	
Firewall throughput[2] (IMIX)	12 Gbps
Firewall throughput[2] (1518B)	24 Gbps
IPSec VPN throughput[2] (IMIX)	8 Gbps
IPSec VPN throughput[2] (1400B)	18 Gbps
Application security performance (TPS#/CPS**)	20 Gbps/7.5 Gbps
Next-generation firewall (TPS#/CPS**)[3]	19 Gbps/4.5 Gbps
Secure Web Access Firewall (CPS**)[4]	4 Gbps

[2] Throughput numbers based on UDP packets and RFC2544 test methodology

[3] Next-generation firewall performance is measured with firewall, application security, and IPS enabled

[4] Secure Web Access firewall performance is measured with firewall, application security, IPS, security intelligence, and URL filtering enabled

TPS Method: throughput performance of average HTTP sessions

** CPS Method: short-lived sessions

Specifications	SRX1600
Advanced threat (CPS**)[5]	2 Gbps
Connections per second (64B)	170,000
SSL connections per second	3,500
Maximum concurrent sessions (IPv4 or IPv6)	2 million
Route table size (RIB/FIB) (IPv4)	2 million/1 million
IPSec VPN tunnels	2,000

HPE Mist Networking WAN Assurance and AI operations

Alternatively, the SRX1600 firewall can be operated and orchestrated through the HPE Mist. HPE Mist delivers unprecedented automation using a combination of artificial intelligence, machine learning algorithms, and data science techniques to save time, maximize IT productivity, and deliver the best experience to digital users.

HPE Mist Networking WAN Assurance is built on the HPE Mist and delivers full life cycle management and operations, including AI-native insights, automated speed tests, dynamic packet capture (dPCAP), anomaly detection, and root cause identification that focuses on end users' experience. For Day 0 and Day 1 operations, HPE Mist Networking WAN Assurance also provides orchestration, administration, and ZTP for the SRX1600. See the HPE Mist Networking WAN Assurance data sheet for more information.

Ordering Information

To order HPE Networking SRX Firewalls, and to access software licensing information, please contact your HPE Networking representative.

About HPE

HPE is a leader in essential enterprise technology, bringing together the power of AI, cloud, and networking to help organizations achieve more. As pioneers of possibility, our innovation and expertise advance the way people live and work. We empower our customers across industries to optimize operational performance, transform data into foresight, and maximize their impact. Unlock your boldest ambitions, with HPE. Discover more at HPE.com.

[5] Advanced threat performance is measured with firewall, application security, IPS, security intelligence, URL filtering, and malware protection enabled

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